toe.

Automatic drum transcription — audio to MIDI with a convolutional neural network.

A drum-source → log-mel → CNN → MIDI pipeline, evaluated on IDMT-SMT-Drums V2.

Problem Statement

Given a drum recording, produce a **symbolic MIDI transcription** that names *what* was hit and *when*.

Useful for practice transcription, music education, search, and analysis — but hand-transcription is slow and error-prone.

THREE TARGET CLASSES

- Kick drum
- Snare drum
- Hi-hat

  All hi-hat articulations (closed, open, pedal) collapsed into a single class — see slide 3.

Motivation — and a scope cut

The project is named for Japanese math-rock band *toe*. Takashi Kashikura's playing leans heavily on **open hi-hat work** — washy, half-pedalled, often sustained across phrases.

The original plan was a 4-class head (kick / snare / HH closed / HH open) so the transcriber could preserve that articulation.

IDMT-SMT-Drums V2 — the only dataset available to us — has almost no open-hat onsets and no sustained articulations at all. The distinction was not learnable from the data we had.

SCOPE DECISION

Collapsed HH / CHH / OHH into one hi-hat class.

Train what the data actually supports; leave the open-hat distinction for future work with richer datasets (slide 16).

Pipeline overview

Audioentersat 44.1 kHz, passes through two active stages, and leaves as a MIDI file. Quantization and notation rendering exist in the repo but sit outside the active runtime (per DECISIONS.md).

01 · SEPARATE

Demucs

Isolates the drum stem from a full mix.

`htdemucs → drums.wav`

→

02 · TRANSCRIBE

DrumCNN

Per-frame 3-class onset detection on a log-mel spectrogram.

`drums.wav → drums.mid`

03 · QUANTIZE (deferred)

16th-note grid

Snaps onsets to MIDI tempo map.

→

04 · RENDER (deferred)

PDF notation

5-line percussion staff via matplotlib.

Related work

CNN-based ADT

Schlüter & Böck (2014) established that convolutional networks on mel-spectrograms outperform handcrafted onset detectors for percussion. Southall et al. extended this to multi-class drum transcription.

Our DrumCNN follows this family: small stacked Conv2d blocks, frame-level multi-label output.

Why CNNs struggle

CNNs are strong at local *pattern matching* inside a short context window but weak at long-range *temporal reasoning*.

A snare lives in milliseconds; a cymbal swell lives in seconds. A 9-frame window sees one but not the other.

Input: log-mel spectrogram

- Audio loaded at `44100` Hz, mono

  Matches training SR; no resampling needed.

- STFT with `hop = 256` samples (≈ 5.8 ms/frame)

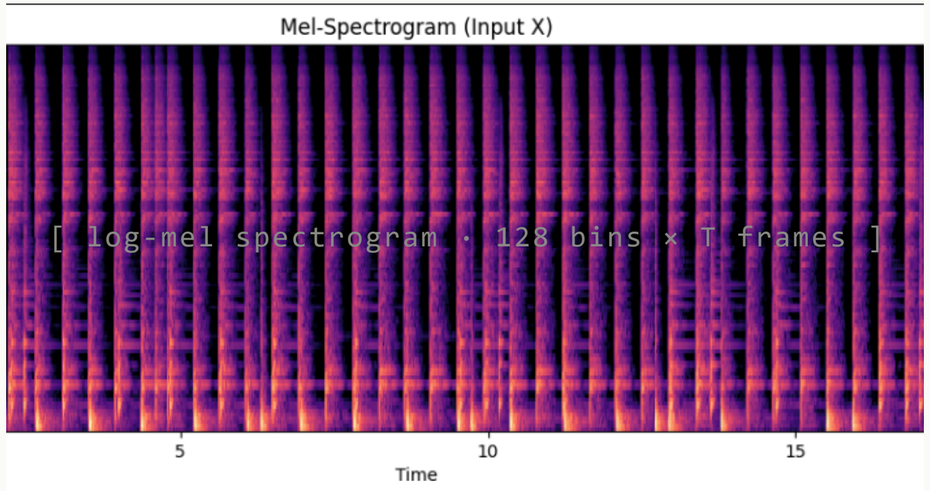
  Setsthetimeresolution ofeverydownstreamdecision.

- `128` mel bins, log-magnitude

  Perceptualfrequencyscale—emphasises low-mid, compresses highs.

- Per-frame `CONTEXT = 13` frame window (~75 ms)

  The temporal receptive field the CNN is allowed to reason over.



128 × 9 is the exact tensor the CNN sees per decision.

PREPROCESSING CRITIQUE

The mel filter is a human-centric lens.
The model uses it to find pixels — not feeling.

DrumCNN — architecture

A small 4-block 2DCNN.Input is a single mel patch; output is a 3 way sigmoid — one probability per drum class per frame.

INPUT	mel patch	(B, 1, 13, 128)
BLOCK 1	Conv2d 1→16, k=3×7 → BN → ReLU → MaxPool(1,2)	(B, 16, 13, 64)
BLOCK 2	Conv2d 16→32, k=3×5 → BN → ReLU → MaxPool(1,2)	(B, 32, 13, 32)
BLOCK 3	Conv2d 32→64, k=3×3 → BN → ReLU → MaxPool(1,2)	(B, 64, 13, 16)
BLOCK 4	Conv2d 64→64, k=3×3 → BN → ReLU → AdaptiveAvgPool(1,1)	(B, 64, 1, 1)
HEAD	Flatten → Linear(64→32) → ReLU → Dropout(0.3) → Linear(32→3)	(B, 3) · sigmoid

Training setup

Dataset

- IDMT-SMT-Drums V2

  Isolated drum recordings with XML onset annotations.

- Frame-level binary labels per class

  DerivedbyaligningXMLonsetstomelframe indices.

Handling imbalance

- `WeightedRandomSampler` — onset frames sampled 10×

  Onsets are only ~2–5% of frames; uniform sampling floods each batch with silence and gradients collapse to all-zero. Re-weighting keeps every mini-batch balanced.

- `BCEWithLogitsLoss, pos_weight = n_neg / n_pos`

  Per-class priors still differ (hi-hats ≫ snares). pos_weight rescales each class's positive loss so missing a rare snare costs as much as missing a common hi-hat.

OPTIMIZER

Adam

`lr = 1e-3`

SCHEDULER

ReduceLROnPlateau

`patience=3, factor=0.5`

INFERENCE

p ≥ 0.6

`175ms inter-onset gap`

CHECKPOINT

Best val loss

`output/drum_cnn.pt`

Evaluation setup

Protocol

- 95 `#MIX` drum loops

  IDMT-SMT-Drums V2 — audio + XML onsets.

- `mir_eval.util.match_events`

  Standard matching for onset-based tasks.

- Tolerance window: `50 ms`

  Convention for drum onset benchmarks.

Metrics

```
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2·P·R / (P + R)
```

Counted per class; aggregated across all 95 files; macro-averaged over the 3 classes.

Results — per-class F1

CLASS	PRECISION	RECALL	F1
Kick	0.946	0.999	**0.972**
Snare	0.878	10.989	**0.930**
Hi-hat	0.9196	0.983	**0.959**
Macro average	**0.908**	**0.998**	**0.951**

Across all three classes, recall is very high (≥ 0.98) — almost every ground-truth onset is found. Precision is the weaker axis; the model produces some false positives, especially on snare. Macro F1 = 0.951.

Failure mode · temporal smearing — not supported

HYPOTHESIS

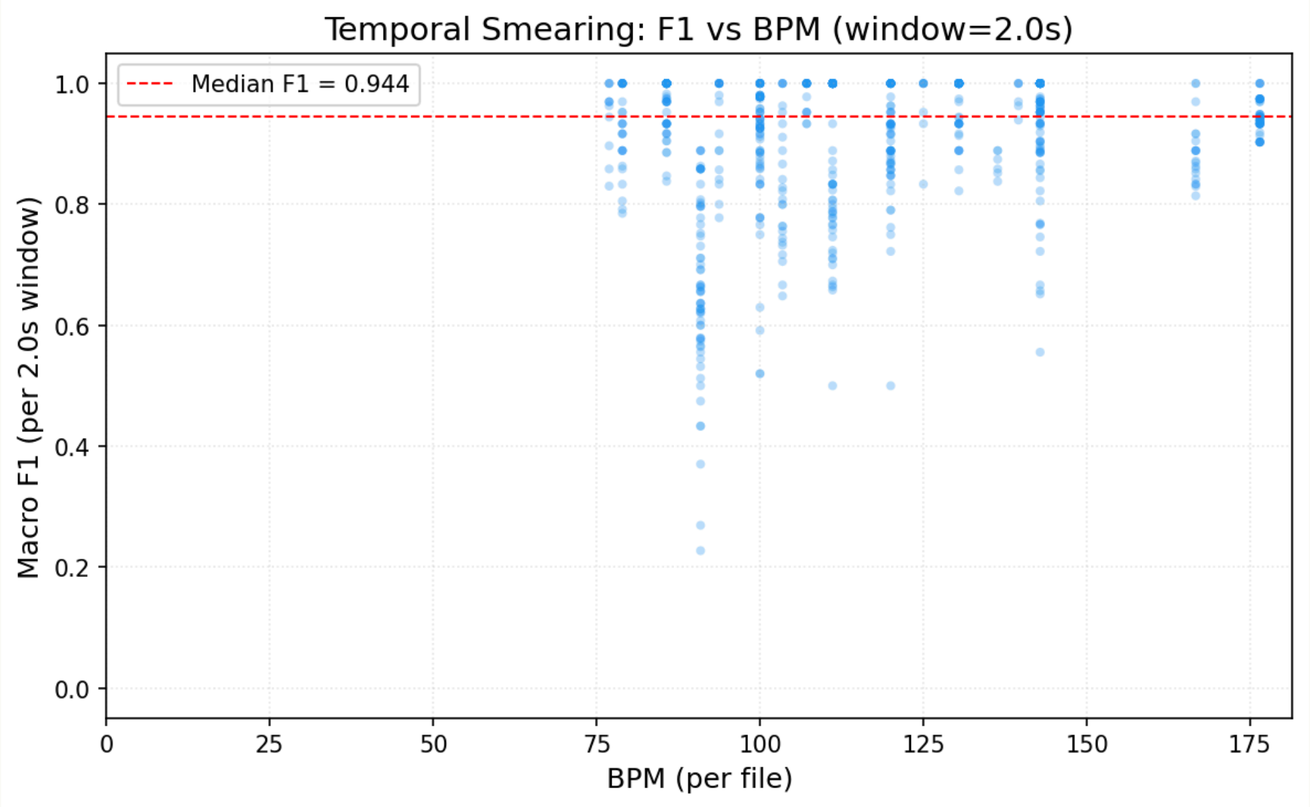
As BPM rises, inter-onset intervals shrink past the STFT hop — adjacent hits should *smear* into one mel patch and F1 should fall.

diagnose.py · 704 windows, 2 s each, split at the median BPM of 111.1.

OBSERVED

```
BPM ≤ 111.1 → → F1 = 0.880  (N=384)
BPM > 111.1     F1 = 0.937  (N=320)
Δ = +0.057
```

F1 is *higher* at faster tempo. The hypothesis does not hold on this dataset.



F1 vs BPM — 704 two-second windows. Contrary to the hypothesis, F1 is slightly *higher* above the median tempo.

Failure mode · spectral masking — not supported

HYPOTHESIS

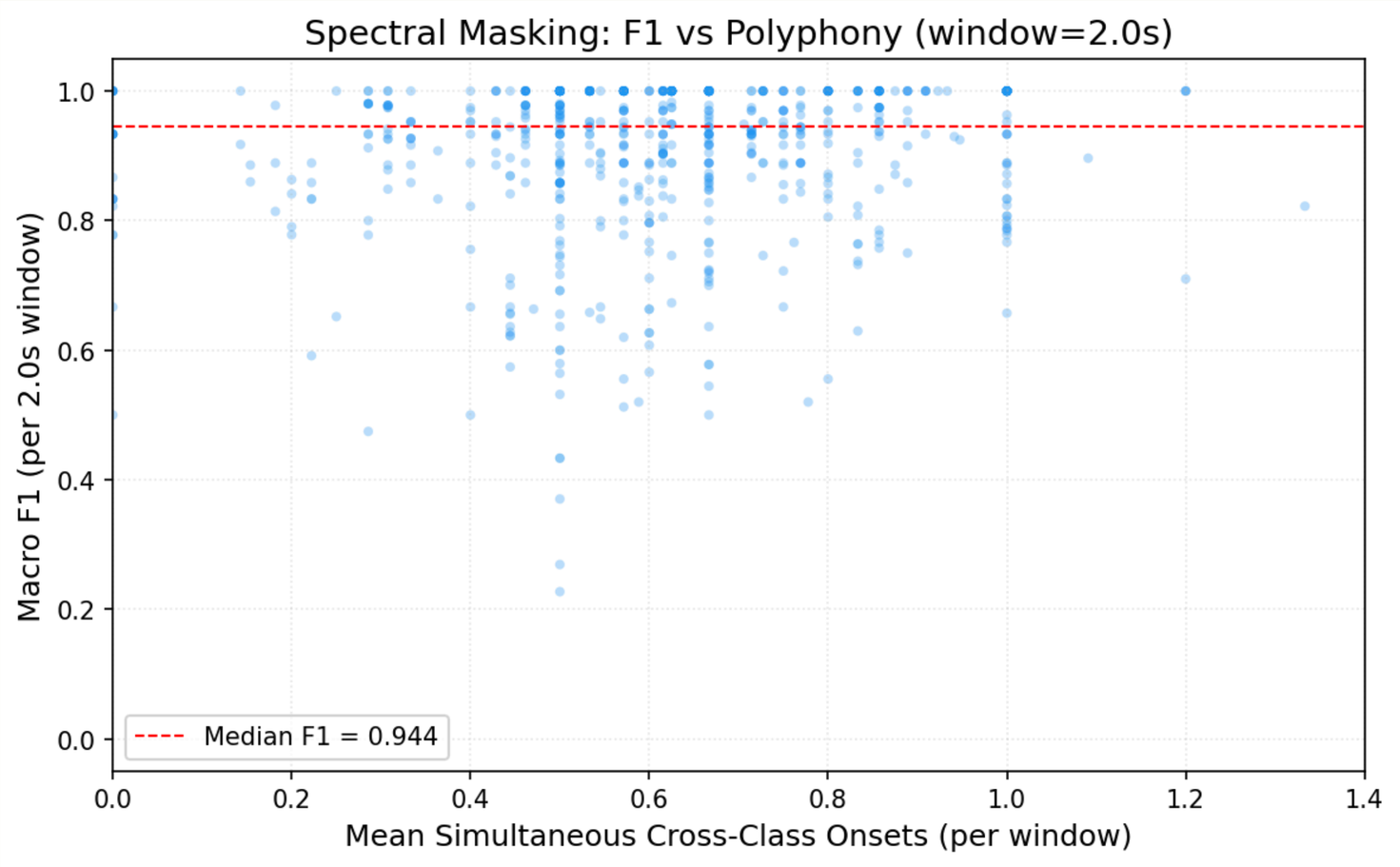
Simultaneous onsets should overlap in frequency — louder broadband events *mask* quieter transients — so F1 should fall with polyphony.

Polyphony = cross-class onset co-occurrence within 50 ms. Split at median 0.6.

OBSERVED

```
poly ≤ 0.6    → F1 = 0.910  (N=361)
poly > 0.6`   → F1 = 0.954  (N=343)
Δ = +0.043
```

F1 is *higher* in denser windows. Masking — if it exists here — is not the dominant factor.



F1 vs polyphony — dense windows perform slightly *better*. See notes.

Reading the result — why the hypotheses failed

0.920

macro F1 across kick, snare, hi-hat on IDMT-SMT-Drums V2

0.996

macro recall - almost every ground-truth onset is found

704

two-second diagnostic windows `analysed by diagnose.py`

0

of the two a-priori failure-mode hypotheses survived contact with the data

Why didn't the failure modes show up?

- IDMT is *uniformly clean* — isolated kits, three rooms, short loops. The frontend is never challenged enough for masking or smearing to dominate.

  Dense windows tend to be the more structured parts of a loop (a groove, not a fill) — which are actually easier to classify

- BPM and polyphony are *confounded* with groove regularity on this dataset. Fast, dense loops are still well behaved pop/rock patterns with large inter-class separation

What this actually tells us

- The bottleneck isn't the CNN architecture — it's the *evaluation surface*.

  0.95 F1 on IDMT doesn't predict performance on the *toe* catalogue; the data distributions aren't comparable.

- To study failure modes honestly we need harder, more realistic data (see next two slides).

Dataset limits — why IDMT doesn't answer the interesting questions

IDMT-SMT-Drums V2 is 95 short loops on three isolated kits.

It is excellent for training a *basic* drum detector — but the genre we care about lives outside its distribution.

Open-hat onsets are almost absent; sustained / half-open articulations aren't represented at all. This is what forced the 3-class scope on slide 3.

Illustrative onset counts across the 95-loop IDMT set; open-hat representation is < 1% of total onsets.

Future work — richer data, stronger models

Extend to real drum datasets

- `ENST-Drums`

  Three professional drummers, multi-track + overheads, phrases and full performances — covers sustained open-hat and ride articulations absent from IDMT.

- `GDB` (Groove MIDI / Drummer Benchmark)

  Real performance datawith humantiming anddynamics — the kind of variability a math-rock transcriber actually has to handle.

The 4-block CNN won't scale

A shallow CNN can memorize isolated IDMT loops but cannot model context at the timescale real drumming operates on.

SOTA ADT systems (ADTOF-RV, MT3-style transcribers, OaF-Drums) pair a CNN frontend with self-attention and recurrent stacks — they can reason over seconds of audio and sustained cymbal events, which is exactly what distinguishes open from closed hi-hat.

OURS

4-block CNN

13-frame window · ~75ms context · IDMT loops only.

`blind to long events`

→

SOTA

CNN + Attention + RNN

Seconds of context · trained on ENST / GDB · models sustain & phrase structure.

`sees the whole phrase`

Conclusion

- A small 4-block CNN on log-mel frames achieves macro F1 = `0.951` on IDMT-SMT-Drums V2 across kick, snare, and hi-hat.

  Recall is near-perfect (0.99); precision is the weaker axis, especially on snare.

- The two a-priori failure-mode hypotheses — temporal smearing at high BPM, spectral masking at high polyphony — were *not supported* by the diagnostic run.

  F1 was slightly *higher* in the fast and dense halves of the dataset. IDMT is too uniformly clean to surface either effect.

- The open hi-hat class was cut deliberately: the dataset contains < 1% open-hat onsets, so the distinction was not learnable.

  Not a modelling failure — a dataset failure.

- The next question is not “a bigger CNN,” it is “a harder dataset.”

  ENSTĐrums and GDB provide the distributional variety needed to retestmasking and smearing honestly; attention + recurrent stacks are the natural follow-on for long-context cymbal articulation.

Questions

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